Jaliya Lakshan Wijayaraja

Embedded Systems & AI Researcher | Educator | Electronic Engineer | Award-Winning Innovator

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ACADEMIC PROFILE

An energetic and self-motivated researcher with a strong engineering foundation in embedded systems, signal processing, and applied AI. Experienced in multidisciplinary studies that bridge research with real-world implementation, with significant research contributions across electronic engineering, artificial intelligence, environmental conservation, disaster management, defense, and biomedical engineering. Recognized for multiple IEEE publications and national innovation awards, and bringing analytical rigor, technical excellence, and a research-driven commitment to technological advancement and societal impact.

EDUCATION QUALIFICATIONS

- **Master of Philosophy in Computer Networks**

  Sri Lanka Institute of Information Technology (2025)

  Thesis title: Elephant Detection and Early Warning Using Seismic Waves.

- **Bachelor of Science of Engineering Honours in Electrical & Electronic Engineering**

  Sri Lanka Institute of Information Technology (2020)

  GPA: 3.62 | Second Class Honours in Upper Division.

- **Professional Graduate Diploma in IT (BCS Level 6)**

  BCS Higher Education Qualification (2017)

CERTIFICATIONS

Machine Learning - *Stanford Online, Coursera (May 2021)* [Credential ID: 5QZ9R9QNPP43]

AWARDS AND SCHOLARSHIPS

- Sri Lanka Institute of Information Technology (SLIIT) Research Excellence Award, 2023 and 2024
- Outstanding Paper Award, International Conference on Advancements in Computing, 2024
- ICT Student Award – Postgraduate Category, Computer Society of Sri Lanka (CSSL National ICT Awards), 2023
- Bronze Award (Postgraduate Category), National ICT Awards – NBQSA Sri Lanka, 2023
- IES Student and Young Professionals Paper Assistance for 49th annual conference of IEEE industrial electronic society.
- Dean's list recognition for academic excellence – BSc Engineering
  [1st Yr 1st Sem (2017) | 2nd Yr 2nd Sem (2018) | 3rd Yr 2nd Sem (2019) | 4th Yr 2nd Sem (2020)]

WORK EXPERIENCE

Lecturer **Feb 2026 - Present**

Department of Industrial Management, University of Kelaniya

- Deliver lectures and conduct assessments for the undergraduate module: Mobile Computing.
- Actively contribute to research in Embedded Systems, IoT, Electronics, and Machine Learning.

Contract R&D Engineer **Sep 2025 - Present**
Independent Subcontractor for CorSci LLC, USA

Subcontracted by CorSci LLC to support the design and development of biomedical device subsystems and related electronic control architectures.

- Design electronic systems and develop firmware for biomedical device subsystems, ensuring reliability and performance optimization.

Co-Founder/ R&D Engineer (Consulting) **Feb 2023 - Present**
Farm's Tech (Pvt) Ltd

Co-founded a technology venture inspired by my MPhil research at SLIIT, focused on delivering advanced engineering solutions for the agricultural sector in Sri Lanka. The venture **secured grant support through the GoviLab AgriTech Accelerator**: a nationally recognized program under Sri Lanka's Inclusive Digital Agriculture Transformation (IDAT) initiative, supported by the Government of Sri Lanka and the Bill & Melinda Gates Foundation.

- Designed and deployed animal repellent systems currently operating at over 240 sites across all districts in Sri Lanka.

- Leading applied research on agentic artificial intelligence for small and medium-scale agricultural enterprises, translating theory into commercially viable products.

Academic Instructor **May 2025 - Feb 2026**
Faculty of Computing, Sri Lanka Institute of Information Technology

- Conduct undergraduate lab sessions in Control Systems Engineering, Embedded Systems, Robotics and Intelligent Systems, IoT, and Hardware Security.

- Supervised final-year research projects as an external mentor in AI and embedded systems

Research Assistant **May 2021 - May 2025**
Faculty of Computing, Sri Lanka Institute of Information Technology

- Worked as a main researcher on interdisciplinary research into human-elephant conflict mitigation, developing AI-based early warning systems using seismic signal processing, embedded engineering, and machine learning.

- Conduct undergraduate lab sessions in Embedded Systems, Robotics and Intelligent Systems, IoT, and Hardware Security.

Resource Person (Visiting Lecturer) **Aug 2022 - Sep 2023**
Metropolitan Campus KDU

- Delivered and developed the specialized course "Certificate in IoT & Big Data" for officers of the Office of the Chief of National Intelligence and Disaster Management Division, Ministry of Defence, Sri Lanka.

- Designed course material focused on IoT and electronics for defense and emergency response applications.

Electronics Engineer (Trainee) **Nov 2019 - Jan 2020**
Centre for Defence Research and Development,
State Ministry of National Security, Home Affairs and Disaster Management.

- Developed a digital voice encryption module; contributed to defence-grade jammer system design (electronic design and PCB layout).

- Designed and prototyped a computer-vision application for a T-55 main battle tank training simulator.

June 2020

- Voluntarily contributed to the design of jammer systems for national interest, focusing on PCB and electronic design.

Engineer Trainee **Nov 2018 - Jan 2019**
Vega Innovations

- Developed the AIGrow Smart Sense Android application for real-time agricultural data monitoring.
- Contributed to developing the AIGrow Sensor Node by designing digital circuits and implementing Node.js-based firmware.

RESEARCH EXPERIENCES

Implementation of an Elephant Monitoring and Early Warning System Using Seismic Waves
[Master of Philosophy research]

This project was undertaken as my MPhil research at SLIIT, where I served as the main researcher in developing engineering solutions to mitigate **human-elephant conflict** through seismic signal analysis. The work involved designing novel analog system architectures for seismic data acquisition, functioning as an interface for embedded systems, and applying machine learning models to classify elephant footfall patterns. This interdisciplinary research bridges embedded systems and wildlife conservation, enabling practical early warning applications. The study was recognized with **two national awards** and has led to **multiple publications** in IEEE Access, a **top 10% Q1-ranked multidisciplinary journal** in engineering and applied sciences.

Evaluation of a Retrieval-Augmented Generation (RAG) Based System for Automated Viva Assessments in STEM Education

This study investigates the use of a Retrieval-Augmented Generation (RAG)–based system as an automated and scalable alternative to traditional oral viva examinations in computing and engineering education. As the principal researcher, I led the design and development of the prototype system, which generates personalized viva questions, analyzes student submissions, and evaluates responses using LLM-driven reasoning. The system was deployed across multiple undergraduate modules and tested with over 200 students, enabling direct comparison with traditional viva evaluations to assess consistency, bias reduction, and pedagogical validity. Early findings demonstrate that the RAG-based approach provides reliable, transparent, and scalable assessment outcomes, supporting its potential integration into higher education assessment practices.

Machine Learning for Flood Forecasting and Analysis in Sri Lanka
[Government Collaboration]

This research aims to develop an accurate flood prediction system for risk management in Sri Lanka using real-time data provided by government agencies. Conducted in collaboration with the Disaster Management Division of the Ministry of Defence, the project leverages machine learning techniques to improve the reliability and responsiveness of flood forecasting systems. The study was **recognized with an Outstanding Paper Award** at ICAC 2024 for its innovation and national relevance.

A Study to Increase the Accuracy of Automobile RPM Sensing from Ignition Pulses of Distributor-Based Ignition Systems.

This research investigates signal processing techniques to enhance the accuracy of engine speed measurements using ignition pulses from distributor-based internal combustion engine systems. Initiated and conducted as the main researcher, the study focused on simulating and evaluating waveform analysis methods for improved RPM sensing. The work was **recognized at IEEE IECON 2023** and was selected for the **IEEE IES Student and Young Professionals Paper Assistance** (IES-SYPA) program

Encrypted Radio for Defence Forces
[Final year research for BSc. Engineering (Hons)]

This research project focused on developing a secure communication system tailored for defense applications in Sri Lanka. It was conducted with the supervision of the Centre for Defence Research and Development, State Ministry of National Security, Home Affairs, and Disaster Management.

PUBLICATIONS

Journal Publications

J. L. Wijayaraja, J. L. Wijekoon, M. Wijesundara and L. J. Mendis Wickramasinghe, "*Toward Long Range Detection of Elephants Using Seismic Signals: A Geophone-Sensor Interface for Embedded Systems*," in **IEEE Access**, vol. 12, pp. 72210-72223, 2024, doi: 10.1109/ACCESS.2024.3401855.

J. L. Wijayaraja, J. L. Wijekoon and M. Wijesundara, "Event Detection and Classification for Long Range Sensing of Elephants Using Seismic Signals," in *IEEE Access*, vol. 13, pp. 158366-158384, 2025, doi: 10.1109/ACCESS.2025.3607383.

Conference Proceedings

J. L. Wijayaraja, N. Abhayasinghe and S. Wellalage, "*Towards Accurate RPM Detection Using Primary Ignition Pulses*," **IECON 2023- 49th Annual Conference of the IEEE Industrial Electronics Society**, Singapore, Singapore, 2023, pp. 1-5, doi: 10.1109/IECON51785.2023.10312638.

J. L. Wijayaraja, N. D. U. Gamage, P. V. N. U. Pallegangoda, N. Priyashantha, W. B. M. R. Weerasekara and P. K. W. Abeygunawardhana, "*Understanding Catchment Impacts: Enhanced Water Level Prediction with LSTM and Explainable AI*," *2024* ***6th International Conference on Advancements in Computing (ICAC)***, Colombo, Sri Lanka, 2024, pp. 480-485, doi: 10.1109/ICAC64487.2024.10851183.

D. Balasingham, S. Samarathunga, A. Bandara, G. G. Arachchige, N. Gamage and J. L. Wijayaraja, "*Enhancing Train Transportation in Sri Lanka: A Smart IOT Based Multi-Subsystem Approach Using MQTT*," ***TENCON 2024 - 2024 IEEE Region 10 Conference (TENCON)***, Singapore, Singapore, 2024, pp. 895-898, doi: 10.1109/TENCON61640.2024.10902958.

L. Jayakody *et al*., "*Forest Fire Prevention and Detection System*," **2025 IEEE 5th International Conference on Power, Electronics and Computer Applications (ICPECA)***,* Shenyang, China, 2025, pp. 748-753, doi: 10.1109/ICPECA63937.2025.10928745.

Working Papers

Evaluating a RAG-Based AI System as an Alternative to Traditional Oral Viva Examinations in STEM Education.
[*Status: Manuscript under preparation*]

A Conversational Agentic AI Framework for Enhancing Operational Excellence in SMEs.
[*Status: Manuscript under preparation*]

KEY SKILLS

- **Embedded Systems Engineering and Firmware Development** - *Microcontroller-based design, real-time interfacing, and embedded C programming*
- **Artificial Intelligence and Machine Learning** - *Applied ML for signal classification, forecasting, agentic AI, retrieval-augmented generation (RAG), and explainable AI applications*
- **Computer Vision and Intelligent Systems** - *Visual recognition, simulator integration, and OpenCV-based automation*
- **Signal Processing and Simulation** - *Temporal and spectral analysis, digital filter design, event detection algorithm design, seismic signal analysis, signal modelling and simulating using MATLAB*
- **Electronic Designing and Simulating** - *Analog/digital circuit design and testing with Proteus and Multisim*
- **PCB Designing** - *Schematic layout and board routing using Eagle CAD*
- **3D CAD Modelling** - *Product casing and prototyping using Autodesk Fusion 360*
- **FPGA & Digital Logic Design** - *VHDL-based implementation for digital control and logic systems*
- **Programming Proficiency** - *C, Python, Java, JavaScript, PHP*
- **Academic & Technical Writing** - *Scientific and academic writing in LaTeX*

LANGUAGE PROFICIENCY

- English: Fluent
- Sinhala: Native

EXTRACURRICULAR ACTIVITIES

- Member of Engineering Research and Innovation Club – Faculty of Engineering, SLIIT
- 1st place in District open Wushu competition - 2014
- The rank of purple belt in the art of Juuoshin Ryu jui-jitsu.
- Purple belt in Shaolin Martial Arts.
- Member of prefects guild – Govt. Science College, Matale.

REFERENCES

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